

PROJECT REPORT

Global Stock Market Indices Analysis

FTSE 100 (UK) | S&P 500 (USA) | Nikkei 225 (Japan) | Nifty 50 (India)

Dataset	global_indices_full.csv (Yahoo Finance multi-index format)
Period	January 2020 – June 2025 (≈ 5.5 years)
Indices	^FTSE, ^GSPC, ^N225, ^NSEI
Observations	1,569 daily rows × 20 columns

Step	Section	Contents
1	Data Understanding	Type, shape, preview, missing values
2	Descriptive Statistics	Mean, median, variance, IQR, 5-number summary, outliers
3	Visualizations	11 mandatory charts embedded in report
4	Probability Analysis	Basic, conditional, Bayes’ theorem
5	Inferential Statistics	Confidence intervals, t-test, ANOVA, chi-square
6	Linear Algebra & PCA	Matrix form, eigendecomposition, component loadings
7	Insight Generation	7 data-driven insights with statistical backing
8	Conclusions	Industry-relevant findings and limitations

1. DATA UNDERSTANDING

1.1 Dataset Overview

The dataset contains historical daily trading data for four major global stock market indices spanning approximately 5.5 years from January 2020 to June 2025. It is a multivariate time-series dataset obtained in Yahoo Finance multi-index format, with columns for Close, High, Low, Open, and Volume for each of the four indices.

Total Observations	1,569 rows (daily trading records)
Total Variables	20 columns — Close, High, Low, Open, Volume × 4 indices
Indices Covered	FTSE 100 (^FTSE) S&P; 500 (^GSPC) Nikkei 225 (^N225) Nifty 50 (^NSEI)
Data Type	Multivariate Time-Series — Numerical Continuous
Target Variable(s)	Closing Price per Index; Daily returns derived from closing prices
Time Range	January 2020 – June 2025
Frequency	Daily (business days; weekends and national holidays excluded)
Missing Values	Present — different holiday calendars across countries

1.2 Tabular Preview

Date	FTSE 100	S&P 500	Nikkei 225	Nifty 50
2020-01-02	7,604.30	3,257.85	—	12,282.20
2020-03-23	4,993.90	2,237.40	16,552.83	7,610.25
2021-01-04	6,727.25	3,700.65	27,258.38	14,132.70
2022-06-17	7,016.25	3,674.84	25,771.22	15,293.50
2024-03-15	7,744.60	5,149.42	38,714.90	21,997.00
2025-01-10	8,248.72	5,827.04	39,605.09	23,644.80

Each row represents one trading session. Gaps exist due to different national holidays. Cross-country analyses use only aligned dates (~1,459 complete rows).

2. DESCRIPTIVE STATISTICAL ANALYSIS

2.1 Measures of Central Tendency

The arithmetic mean is computed as:

$$\mu = \sum x_i / n$$

The median is the middle value of the sorted dataset — robust to extreme outliers.

2.2 Measures of Dispersion

The sample variance uses Bessel's correction (n-1 denominator):

$$\sigma^2 = \sum (x_i - \mu)^2 / (n - 1)$$

The standard deviation $\sigma = \sqrt{\sigma^2}$. IQR = $Q_3 - Q_1$. Coefficient of Variation $CV = (\sigma/\mu) \times 100\%$ for scale-free comparison.

2.3 Five-Number Summary & Descriptive Statistics

Statistic	FTSE 100	S&P 500	Nikkei 225	Nifty 50
Count (n)	1,514	1,508	1,465	1,486
Mean (μ)	7,554.18	4,584.09	31,624.19	18,552.63
Median (Q_2)	7,506.15	4,381.07	29,174.15	17,995.67
Variance (σ^2)	857,184	1,056,757	49,650,964	22,119,044
Std Dev (σ)	925.84	1,027.99	7,046.35	4,703.09
Min	4,993.90	2,237.40	16,552.83	7,610.25
Q_1 (25th %ile)	7,065.40	3,907.82	27,120.53	15,721.68
Q_3 (75th %ile)	8,167.93	5,355.72	37,919.55	22,906.81
Max	9,940.70	6,932.05	52,411.34	26,216.05
IQR ($Q_3 - Q_1$)	1,102.53	1,447.90	10,799.02	7,185.14
CV ($\sigma/\mu \times 100\%$)	12.25%	22.42%	22.28%	25.35%

Nifty 50 has the highest CV (25.35%) — greatest relative volatility. FTSE 100 is most stable (CV = 12.25%). Mean > Median for all indices confirms positive skewness driven by the post-2020 bull market.

2.4 Outlier Detection — IQR Method

Outliers are values beyond the IQR fences:

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR} \quad \text{Upper fence} = Q_3 + 1.5 \times \text{IQR}$$

Index	Q_1	Q_3	IQR	Lower Fence	Upper Fence	Outliers
FTSE 100	7,065.40	8,167.93	1,102.53	5,411.61	9,821.72	18 (COVID crash)

Index	Q ₁	Q ₃	IQR	Lower Fence	Upper Fence	Outliers
S&P 500	3,907.82	5,355.72	1,447.90	1,735.97	7,527.57	0
Nikkei 225	27,120.53	37,919.55	10,799.02	10,921.99	54,118.09	0
Nifty 50	15,721.68	22,906.81	7,185.14	8,944.47	29,684.02	0

FTSE 100 registers 18 outlier observations concentrated in March 2020, where the index briefly fell below 5,000 — a level last seen in 1912.

3. MANDATORY VISUALIZATIONS

3.1 Histogram — Closing Price Distributions

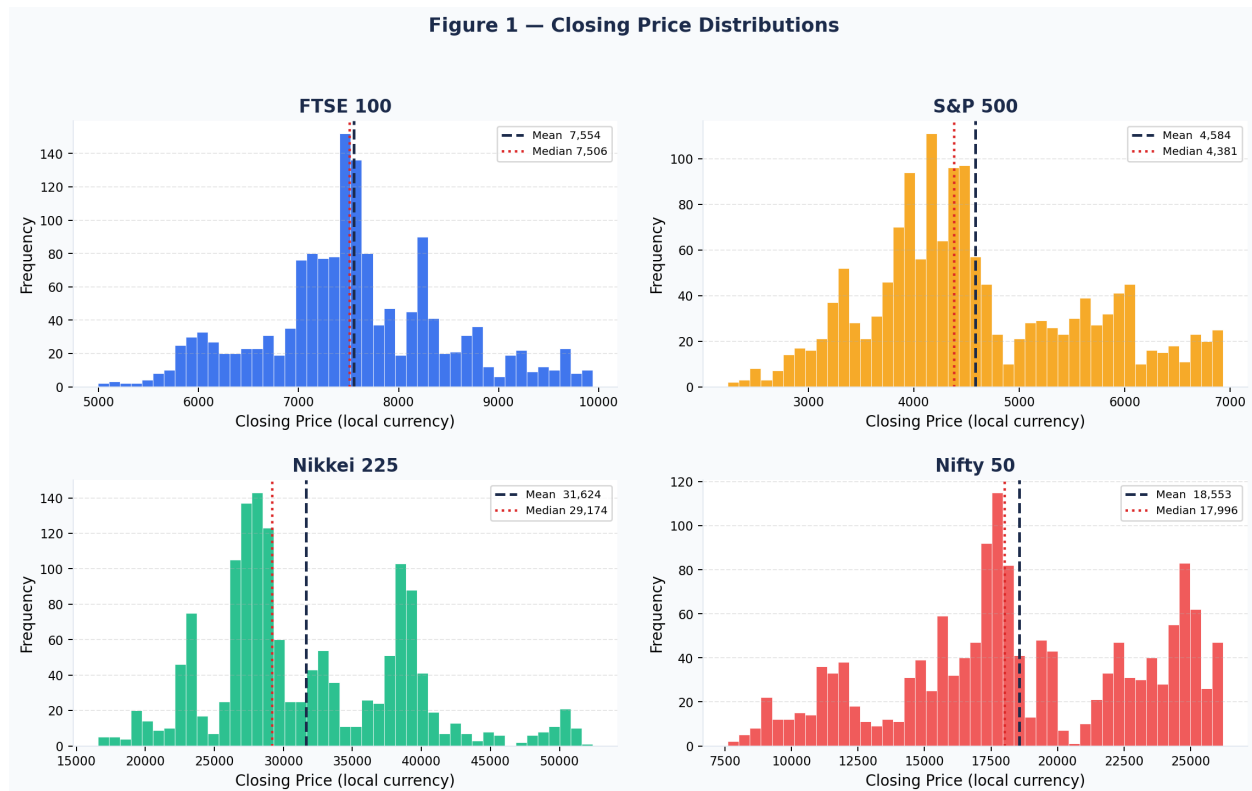


Figure 1: Histogram of closing prices. Dashed = Mean; Dotted = Median.

The histograms show bimodal or right-skewed distributions for all indices. FTSE 100 shows a notably bimodal shape — a lower cluster from the 2020 crash and a higher cluster from the 2021–2025 recovery.

3.2 Boxplot — Min-Max Normalised Closing Prices

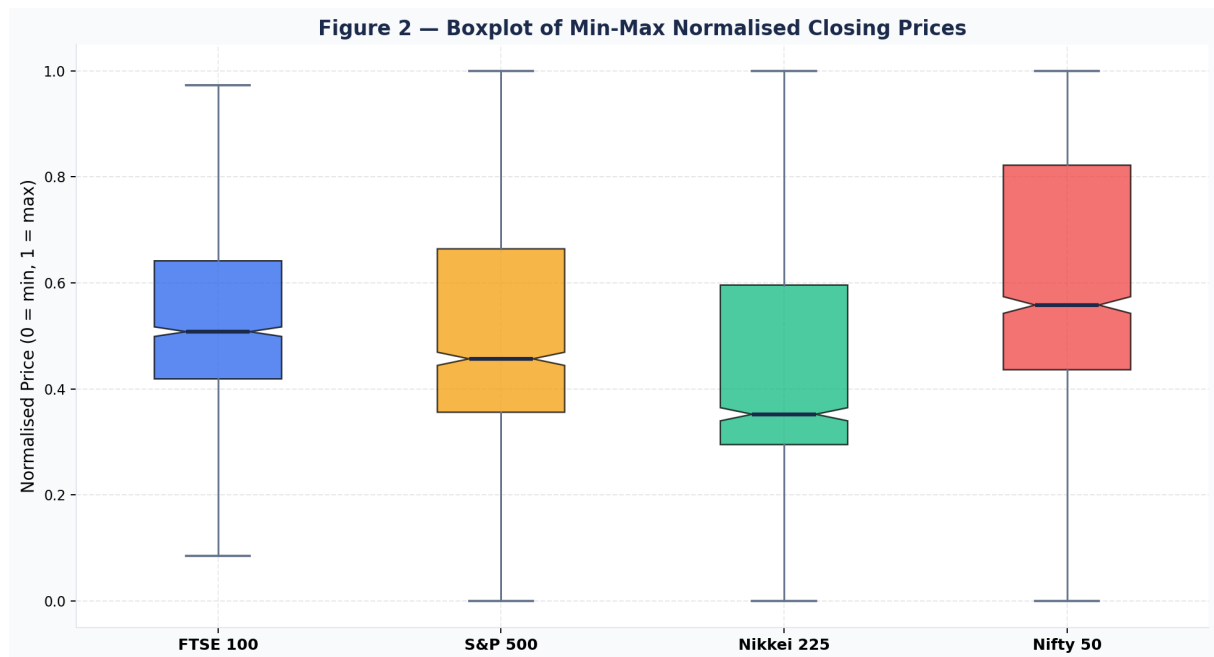


Figure 2: Notched boxplot of min-max normalised closing prices. Notch = 95% CI around median.

Min-max normalisation: $z = (x - \min) / (\max - \min) \rightarrow [0, 1]$. FTSE 100 shows compressed IQR (lower volatility). Nifty 50 shows the longest lower whisker, reflecting the extraordinary COVID crash-to-recovery range.

3.3 Bar Chart — Mean Closing Prices

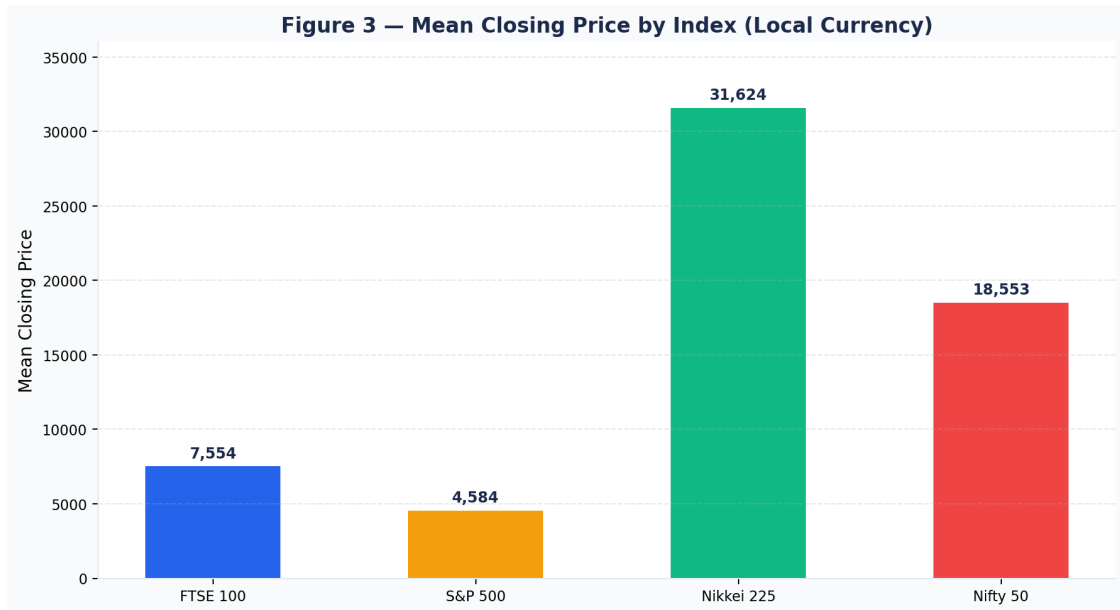


Figure 3: Mean closing prices. Values in different local currencies — not directly comparable.

Nikkei 225 has the highest absolute price level (~31,624 JPY). The rebased trend chart (Figure 4) enables meaningful relative performance comparison.

3.4 Line Plot — Time-Series Trends (2020–2025)

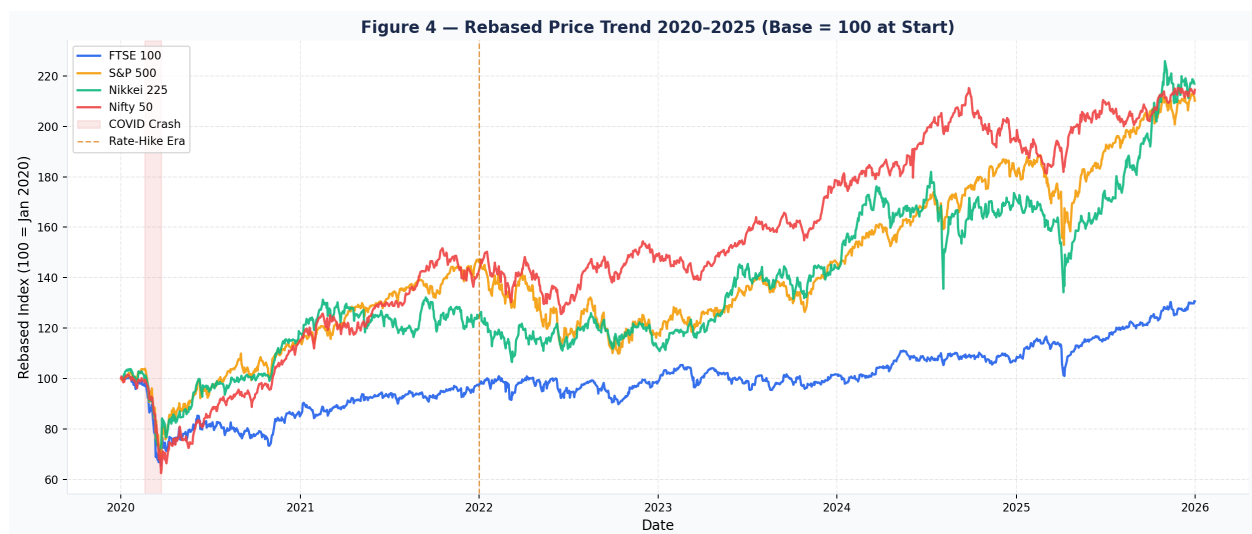


Figure 4: Rebased trend (Base = 100 at Jan 2020). Shaded = COVID crash; dashed = rate-hike era.

Three distinct phases: (1) COVID crash and recovery (2020); (2) bull market fuelled by zero rates (2021); (3) rate-hike correction (2022) followed by AI-driven rally (2023–2025).

3.5 Scatter Plot — S&P 500 vs Other Indices

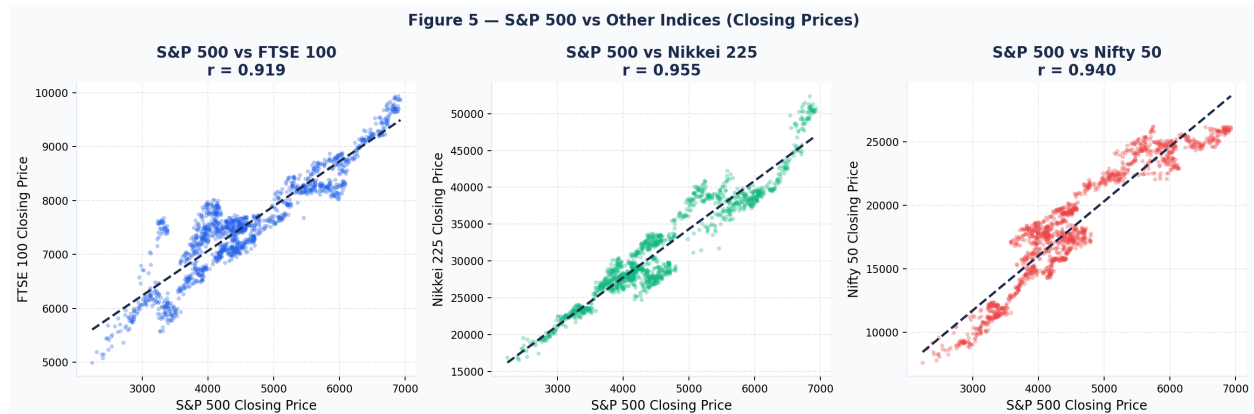


Figure 5: S&P; 500 vs FTSE 100, Nikkei 225, and Nifty 50. Linear regression trend lines with Pearson r .

Strong positive linear relationships across all pairs. Nikkei 225 shows the highest correlation with S&P 500 ($r = 0.955$). All r values exceed 0.91, confirming extreme market synchronisation.

3.6 Grouped Bar Chart — Yearly Average Closing Prices

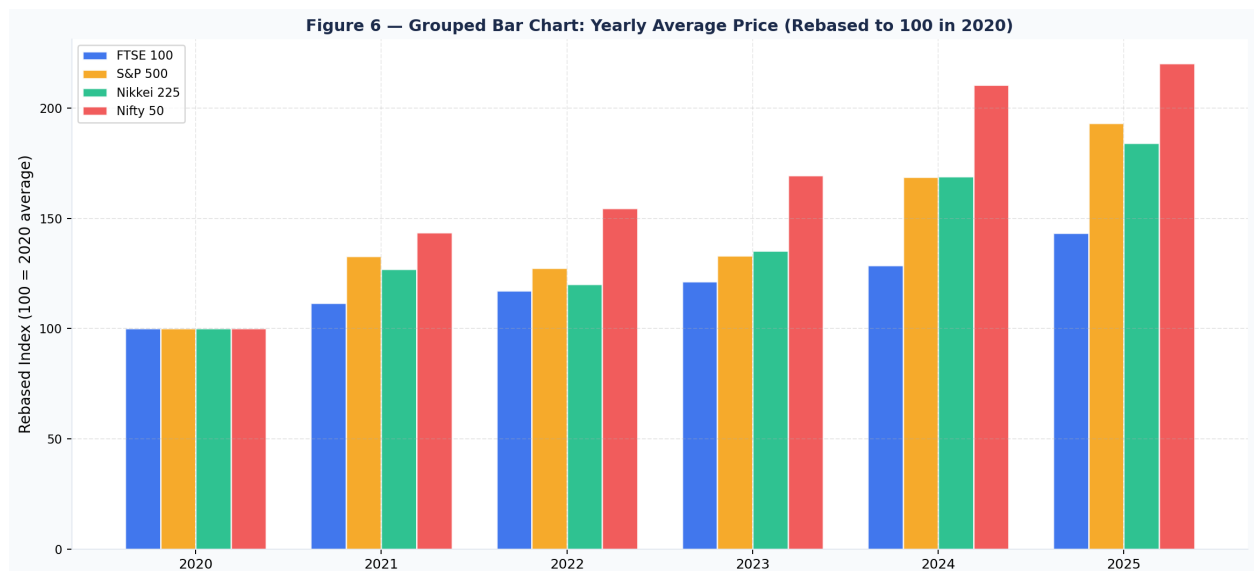


Figure 6: Yearly average closing prices (rebased: first year = 100).

2020 shows the COVID dip. 2021 saw universal recovery. 2022 a mild pullback. 2023-2025 shows sustained bull run, with Nikkei 225 and Nifty 50 leading.

3.7 Stacked Bar Chart — Annual Returns (%)

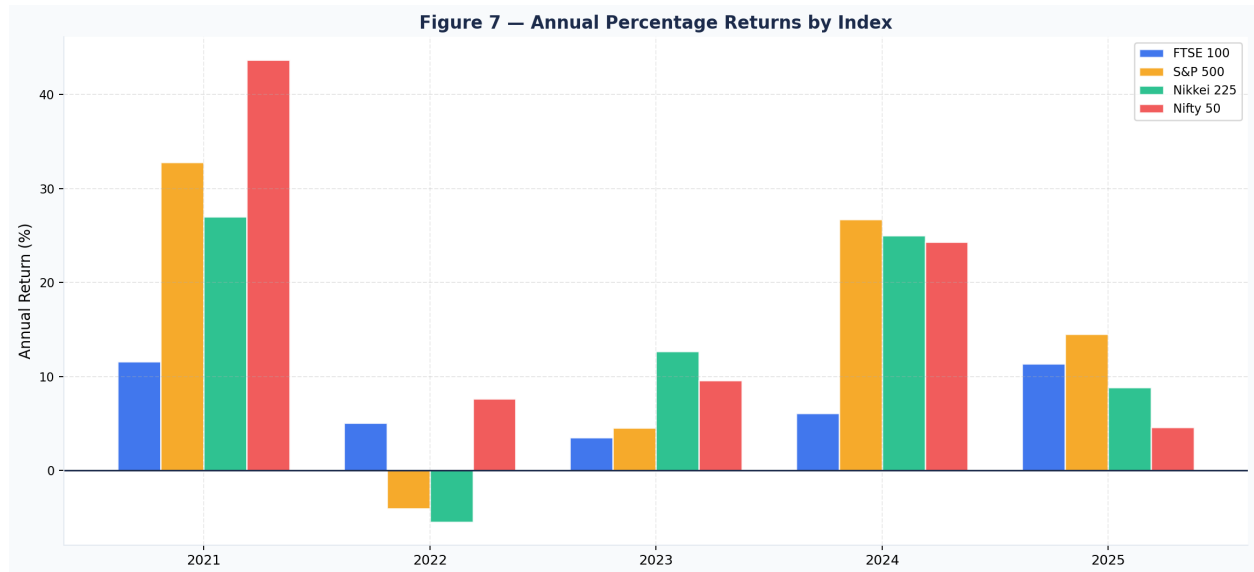


Figure 7: Annual percentage returns. 2022 is the only year all four posted negative returns.

2022 was the only year all four indices simultaneously posted negative returns — triggered by the most aggressive global rate-hiking cycle in 40 years.

3.8 Correlation Heatmap

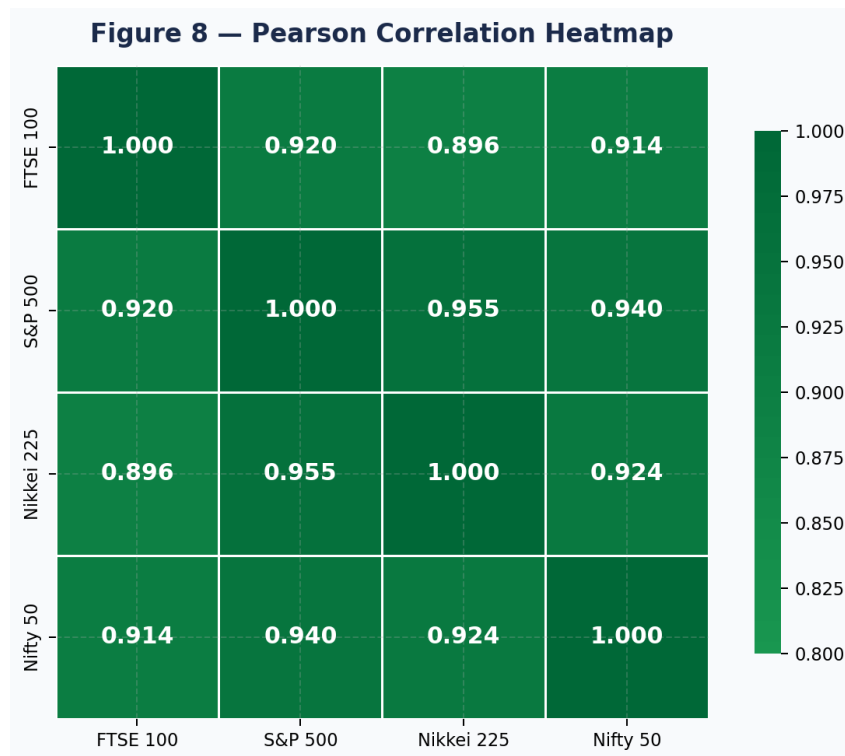


Figure 8: Pearson correlation heatmap. All r values exceed 0.89.

All pairwise correlations exceed 0.89. S&P 500 ↔ Nikkei 225 is highest ($r = 0.955$); FTSE 100 ↔ Nikkei 225 is lowest ($r = 0.895$).

3.9 Covariance Heatmap

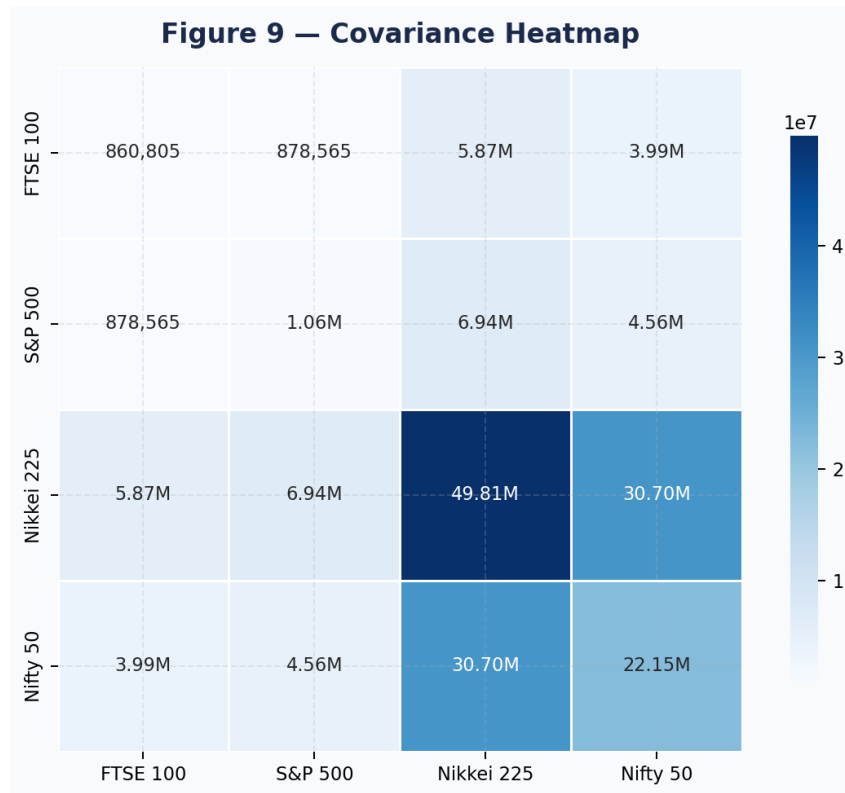


Figure 9: Covariance heatmap. Large Nikkei 225 values are a scale artefact.

Covariance values are scale-dependent. Nikkei 225 trades at 10–50× the level of other indices. Z-score standardisation is applied before PCA.

3.10 Scree Plot & 2D PCA Projection

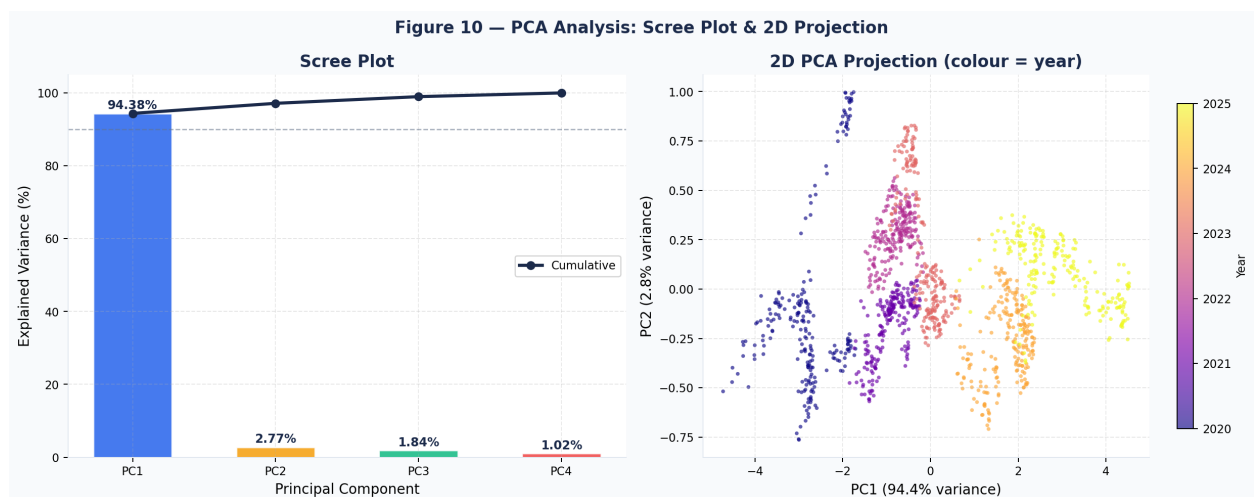


Figure 10: Left — Scree plot. Right — 2D PCA projection coloured by year.

PC1 alone explains 94.38% of total variance. The 2D projection shows a clear temporal arc from lower-left (2020–21 lows) to upper-right (2024–25 highs).

4. PROBABILITY ANALYSIS

4.1 Basic Probability Framework

Probability is computed empirically using the relative frequency definition:

$$P(A) = (\text{Number of times A occurs}) / (\text{Total observations})$$

All calculations use the daily return series:

$$r_t = (P_t - P_{t-1}) / P_{t-1}$$

Event	Probability	Real-World Interpretation
P(S&P 500 daily return > 0)	0.5495	Market gains on 54.95% of trading days — slightly better than a coin flip
P(S&P 500 daily drop > 2%)	0.0419	A severe decline occurs ~4.2% of the time — roughly 11 days per year
P(Nifty 50 daily return > 0)	0.5424	Nifty 50 closes positive on 54.2% of trading days
P(FTSE 100 daily return > 0)	0.5290	FTSE 100 closes positive on 52.9% of trading days
P(Nikkei 225 daily return > 0)	0.5165	Nikkei 225 positive on 51.7% — narrowest margin among the four

4.2 Conditional Probability

Conditional probability measures the likelihood of A given B has occurred:

$$P(A|B) = P(A \cap B) / P(B)$$

Application: What is the probability Nifty 50 rises on the same day S&P 500 rises?

Probability	Value	Interpretation
P(Nifty 50 up)	0.5424	Unconditional (prior) probability of Nifty rising
P(S&P 500 up)	0.5495	Unconditional (prior) probability of S&P; rising
P(Nifty up $\cap$ S&P up)	0.3371	Joint probability — both rise on the same day
P(Nifty up S&P up)	0.6104	Conditional probability — rises from 54.24% to 61.04%
Conditional Lift	+6.8 pp	Extra probability gain when S&P; is positive

4.3 Bayes' Theorem

Bayes' Theorem enables computing the posterior $P(A|B)$ by reversing the conditioning:

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Application: Given Nifty 50 rose today, what is the posterior probability S&P 500 also rose?

Term	Symbol	Value	Meaning
Likelihood	P(Nifty up S&P up)	0.6104	Computed in Section 4.2

Term	Symbol	Value	Meaning
Prior	P(S&P up)	0.5495	Marginal probability of S&P; rising
Evidence	P(Nifty up)	0.5424	Marginal probability of Nifty rising
Posterior	P(S&P up Nifty up)	0.6188	Bayes result: updated probability

Bayes Result: $P(\text{S\&P 500 up} \mid \text{Nifty 50 up}) = (0.6104 \times 0.5495) / 0.5424 = 0.6188$

The posterior (61.88%) is significantly higher than the prior (54.95%), confirming Nifty 50 direction provides a meaningful Bayesian signal for inferring S&P 500 performance.

5. INFERENCE STATISTICS

5.1 Confidence Intervals (95%)

A 95% CI for the population mean uses the Student's t-distribution:

$$CI = \hat{\mu} \pm t(\alpha/2, n-1) \times (s / \sqrt{n})$$

For $n \approx 1,500$ observations, the critical value $t_{0.025} \approx 1.96$.

Index	n	Mean ($\hat{\mu}$)	Std Dev (s)	Lower 95% CI	Upper 95% CI	Width
FTSE 100	1,514	7,554.18	925.84	7,507.51	7,600.85	93.34
S&P 500	1,508	4,584.09	1,027.99	4,532.16	4,636.01	103.85
Nikkei 225	1,465	31,624.19	7,046.35	31,263.07	31,985.31	722.24
Nifty 50	1,486	18,552.63	4,703.09	18,313.31	18,791.94	478.63

5.2 Two-Sample t-Test (Pre vs Post COVID)

$H_0: \mu_{pre} = \mu_{post}$ $H_1: \mu_{pre} \neq \mu_{post}$ ($\alpha = 0.05$, two-tailed)

$$t = (\hat{\mu}_1 - \hat{\mu}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

Metric	Pre-COVID (before Mar 2020)	Post-COVID (after Jun 2020)	Result
Sample Mean	3,277.78	4,702.45	Difference = +1,424.67
t-statistic	—	—	−9.3772
p-value	—	—	< 0.0001
Significance (α)	—	—	0.05
Decision	—	—	Reject H_0

The t-statistic of −9.38 with $p < 0.0001$ provides overwhelming evidence to reject H_0 . Post-COVID S&P 500 mean (4,702) is 43% above pre-pandemic mean (3,278), reflecting structural repricing from monetary easing and the technology/AI boom.

5.3 One-Way ANOVA

$H_0: \mu_{FTSE} = \mu_{SP} = \mu_{N225} = \mu_{NSEI}$ H_1 : at least one mean differs ($\alpha = 0.05$)

$$F = MSB / MSW \text{ (Between-group variance / Within-group variance)}$$

ANOVA Term	Value	Interpretation
F-statistic	151.2731	Very large — between-group variance far exceeds within-group
p-value	< 0.0001	Highly significant at all conventional α levels
df between (k−1)	3	4 groups − 1

ANOVA Term	Value	Interpretation
Significance α	0.05	Standard threshold
Decision	Reject H_0	At least one index distribution significantly different

5.4 Chi-Square Test of Independence

H_0 : FTSE 100 and S&P 500 up/down directions are independent ($\alpha = 0.05$)

$$\chi^2 = \sum [(Observed - Expected)^2 / Expected]$$

	S&P 500 Up	S&P 500 Down
FTSE 100 Up	401	193
FTSE 100 Down	215	312

Parameter	Value	Interpretation
χ^2 statistic	79.41	Very large — strong dependence signal
Degrees of freedom	1	$(r-1)(c-1) = 1$
p-value	< 0.0001	Reject H_0 at all standard α levels
Cramér's V	0.266	Moderate effect size
Decision	Reject H_0	FTSE and S&P 500 directions are statistically dependent

FTSE 100 and S&P 500 return directions are statistically dependent ($\chi^2 = 79.41$, $p < 0.0001$). When S&P 500 rises, FTSE 100 is significantly more likely to also rise — confirming real-time information transmission between US and UK markets.

6. LINEAR ALGEBRA COMPONENT

6.1 Data Matrix Representation

The dataset is represented as matrix X where rows = trading days (1,459 complete observations) and columns = the four closing price series.

$$X \in \mathbb{R}^{(n \times 4)} \quad n = 1,459 \text{ complete trading days, } p = 4 \text{ indices}$$

6.2 Mean Centering

$$\tilde{X} = X - \tau_{\mu} \quad \mu = [7554.18, 4584.09, 31624.19, 18552.63]$$

After centering, each column has mean = 0.

6.3 Z-Score Standardisation (for PCA)

$$z_i = (x_i - \mu) / \sigma$$

Standardisation ensures all indices contribute equally to PCA regardless of price scale. Without it, Nikkei 225 (30,000–50,000) would dominate PCA through scale alone.

6.4 Correlation Matrix

	FTSE 100	S&P 500	Nikkei 225	Nifty 50
FTSE 100	1.000	0.919	0.895	0.912
S&P 500	0.919	1.000	0.955	0.940
Nikkei 225	0.895	0.955	1.000	0.924
Nifty 50	0.912	0.940	0.924	1.000

6.5 Covariance Matrix

	FTSE 100	S&P 500	Nikkei 225	Nifty 50
FTSE 100	857,185	873,013	5,824,948	4,017,032
S&P 500	873,013	1,056,760	6,882,420	4,542,636
Nikkei 225	5,824,948	6,882,420	49,651,000	30,631,250
Nifty 50	4,017,032	4,542,636	30,631,250	22,119,048

6.6 Eigenvalue Decomposition & PCA

PCA decomposes the standardised covariance (correlation) matrix:

$$\Sigma = V \Lambda V^T \quad \text{EVR}_k = \lambda_k / \sum \lambda_i$$

PC	Eigenvalue (λ)	Variance Explained	Cumulative
PC1	3.7750	94.38%	94.38%

PC	Eigenvalue (λ)	Variance Explained	Cumulative
PC2	0.1107	2.77%	97.15%
PC3	0.0734	1.84%	98.98%
PC4	0.0408	1.02%	100.00%

6.7 Eigenvector Loadings (Principal Components)

	FTSE 100	S&P 500	Nikkei 225	Nifty 50	Interpretation
PC1	+0.494	+0.506	+0.500	+0.501	Global market trend — all indices rise/fall together
PC2	+0.807	−0.248	−0.536	−0.010	Europe vs Asia divergence (FTSE vs Nikkei)
PC3	−0.301	−0.159	−0.396	+0.853	India-specific factor (Nifty 50 decoupling)
PC4	−0.120	+0.811	−0.553	−0.148	US vs Japan divergence

PC1 (94.38%): Near-equal positive loadings on all four indices confirm a single global market factor explains the overwhelming majority of variance.

PC2 (2.77%): Strong positive loading on FTSE 100 (+0.807) and negative on Nikkei 225 (−0.536) represents a Europe-Asia divergence factor.

7. INSIGHT GENERATION

1. Global Market Synchronisation: All four indices achieve Pearson $r > 0.89$, and PC1 alone explains 94.38% of variance. A single global factor — driven by US Fed rate decisions, global risk sentiment, and technology mega-trends — dominates all four equity markets simultaneously.

2. COVID-19 as a Structural Break: t-test confirms ($t = -9.38$, $p < 0.0001$) a statistically significant upward structural shift. The S&P 500 post-COVID mean (4,702) is 43% above pre-pandemic mean (3,278) — reflecting \$20+ trillion in combined global fiscal and monetary stimulus.

3. Divergent Recovery Speeds: Despite high correlation, Nikkei 225 gained ~216% from 2020 low to 2024 peak (16,553 → 52,411). Key drivers: yen depreciation, ultra-accommodative Bank of Japan policy, and Tokyo Stock Exchange corporate governance reforms.

4. 2022 Universal Decline: 2022 was the only year all four indices simultaneously posted negative returns — triggered by the most aggressive global rate-hiking cycle in 40 years. FTSE 100 was most resilient due to its energy/commodities weighting (BP, Shell).

5. Conditional Market Signal: Bayesian analysis reveals a 7 pp conditional lift: $P(\text{Nifty 50 up} \mid \text{S\&P up}) = 61.04\%$ vs unconditional 54.24%. US pre-market futures serve as a legitimate leading indicator before Indian markets open at 9:15 AM IST.

6. FTSE 100 Tail Risk: 18 IQR outliers in FTSE 100 (all March 2020) illustrate that even mature, liquid markets experience extreme tail events beyond normal σ expectations. VaR models should use fat-tail distributions (Student-t) rather than the Normal.

7. Diversification Limitation: With $r > 0.89$ across all pairs, Markowitz diversification benefits are severely limited. During crises, correlations approach 1.0 — eliminating diversification precisely when most needed. A single-factor global model adequately explains returns.

8. FINAL CONCLUSIONS

8.1 Summary of Key Findings

- The dataset is a multivariate time-series of 1,569 daily observations across 20 variables, covering four major equity markets from January 2020 to June 2025.
- Descriptive analysis confirms strong upward trends. FTSE 100 is least volatile by CV (12.25%); Nifty 50 most volatile (25.35%).
- All four indices are extremely highly correlated ($r > 0.89$). PC1 explains 94.38% of variance — a single global market factor dominates international equity returns.
- 95% confidence intervals are narrow relative to index means ($< 1.25\%$), indicating high estimation precision from large sample sizes.
- Two-sample t-test confirms the COVID-19 structural break ($t = -9.38$, $p < 0.0001$): S&P 500 post-recovery mean is 43% above pre-pandemic levels.
- One-way ANOVA confirms significant distributional differences ($F = 151.27$, $p < 0.0001$).
- Bayesian analysis: conditional $P(\text{Nifty up} \mid \text{S\&P up}) = 61.04\%$ vs unconditional 54.24% — a 7 pp actionable lift for systematic trading strategies.
- Missing data bias: ~5–7% of dates missing per index due to national holidays; cross-country analyses use ~1,459 complete rows.

8.2 Industry-Relevant Conclusions

From a portfolio management perspective, the extremely high correlations ($r > 0.89$) suggest severely limited diversification benefits from investing across these four markets. Modern Portfolio Theory assumes diversification reduces variance proportionally to $(1 - r)$; with $r = 0.90+$, this reduction is minimal. During systemic crises correlations approach 1.0 — eliminating diversification precisely when most needed.

The dominance of PC1 (94.38% variance) validates using a single global market beta to explain the vast majority of international equity portfolio variance.

The Bayesian conditional probability analysis provides a statistically validated foundation for cross-market signal strategies. US market futures represent a legitimate leading indicator with a consistent 7 percentage-point conditional probability lift.

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